1️⃣ Competency 5: Using definitions, rules, and formal systems (Highest relevance)

Definition:

> Understand and utilise constructs based on definitions, rules and formal systems 🗎 employ

🔍 How it appears in the task

- The problem states:
  - A circular space
  - Three door wings **divide it into three equal sectors** 🗎 revolving-door-qs1
- Key mathematical rules:
  - Total angle of a circle = 360°
  - Three equal sectors ⇒ each angle = 360° ÷ 3

📌 Core insight

The task fundamentally requires:

- Applying **basic geometric definitions** (circle, central angle)
- Using the **equal partition rule**

👉 This is a typical case of:

> "Deriving results directly from mathematical definitions"

✔ Therefore, this is the most central *employ* competency

2️⃣ Competency 1: Performing a simple calculation

Definition:

> Perform a simple calculation 🗎 employ

🔍 How it appears in the task

Calculation involved:

- 360° ÷ 3 = 120°

📌 Characteristics

- No complex reasoning required
- Direct numerical computation

👉 A basic but essential skill

3️⃣ Competency 7: Working with geometric representations

Definition:

> Manipulate … geometric representations employ

🔍 How it appears in the task

- The diagram (page 1) shows:
  - A circular structure
  - Three rotating wings dividing the space

📌 Student requirements

Students must:

- Interpret the diagram
- Recognize:
  - central angles
  - equal partitioning

👉 This reflects:

> "Extracting meaning from geometric representations"

4️⃣ Competency 6: Extracting information from diagrams

Definition:

> Develop diagrams… and extracting mathematical information employ

🔍 How it appears in the task

Students need to identify:

- The circle is divided into three equal sectors
- The angle between two wings is a central angle

📌 Key point

The diagram is not decorative—it is:

- a **carrier of mathematical information**

5️⃣ Competency 10: Using multiple representations

Definition:

> Make sense of… a variety of representations employ

🔍 How it appears in the task

Two representations are involved:

- Verbal description ("three equal sectors")
- Geometric diagram

📌 Requirement

Students must connect:

> language ↔ geometric structure

6️⃣ Competency 12: Using a multi-step procedure

Definition:

> Use a multi-step procedure employ

🔍 How it appears in the task

Even though simple, it involves steps:

1. Recognize the shape is a circle
2. Identify it is divided into three equal parts
3. Compute the angle

👉 A basic multi-step reasoning process

1️⃣ 能力5：运用定义、规则与形式系统（最高相关）

对应条目：

> Understand and utilise constructs based on definitions, rules and formal systems employ

🔍 题目体现

- 题目给出：
  - 圆形空间
  - 三个门翼 **均分为三个扇形**
- 本质规则：
  - **圆周角总和 = 360°**
  - 三等分 ⇒ 每个扇形角度 = 360° ÷ 3

📌 本质分析

该题核心是：

- 运用**几何基本定义（圆、扇形、角度）**
- 利用**均分规则**

👉 属于典型的：

> "基于数学定义直接推导结果"

✔ 因此这是最核心的 employ 能力

2️⃣ 能力1：进行简单计算

对应条目：

> Perform a simple calculation employ

🔍 题目体现

计算过程：

- 360° ÷ 3 = 120°

📌 特点

- 无复杂推导
- 直接数值计算

👉 属于基础运算能力，但不可或缺

3️⃣ 能力7：处理几何表示

对应条目：

> Manipulate … geometric representations employ

🔍 题目体现

- 图中（第1页）展示：
  - 三个门翼在不同位置
  - 圆形空间被分割

📌 学生需要：

- 理解图形结构
- 识别：
  - 中心角
  - 等分关系

👉 属于：

> “从几何图中提取数学信息”

4️⃣ 能力6：从图形中提取信息

对应条目：

> Develop diagrams… and extracting mathematical information employ

🔍 题目体现

学生需要从图中识别：

- 三个扇区完全相等
- 门翼间的夹角就是中心角

📌 关键点

图形不是装饰，而是：

- 信息载体（equal sectors）

5️⃣ 能力10：理解并使用多种表征

对应条目：

> Make sense of… a variety of representations employ

🔍 题目体现

涉及两种表征：

- 文字描述（three equal sectors）
- 图形表示（圆 + 三翼）

📌 要求

学生需将：

> 语言信息 ↔ 几何结构

建立联系

6️⃣ 能力12：多步骤求解

对应条目：

> Use a multi-step procedure employ

🔍 题目体现

虽然简单，但隐含步骤：

1. 识别是圆
2. 确认被三等分
3. 计算角度

👉 属于简单的多步推理